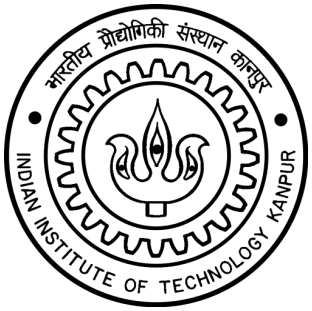


INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — RADIO ASTRONOMY LABORATORY

Experiment 2: RF Components: Characterization and Measurement

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Name: A.Aditya

Roll Number: 261310601

Programme: PhD

Department: SPASE

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1 Aim

To understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

2 Objective

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or pass- band.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(e) To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.

(f) To estimate the input reflection coefficient and return loss of the filter.

3 Material required

The physical apparatus and equipment required to carry out this experiment are listed below :

- RF Signal Generators
- Spectrum Analyser
- Digital Storage Oscilloscope (DSO)
- Filter (Device Under Test)
- Mixer (Device Under Test)
- Two-Way Symmetric Power Splitter/Combiner (Device Under Test)
- Low-pass filter
- RF Tee Connector
- Open-ended Coaxial Stub of physical length L
- Short Reference Coaxial Cable
- High-precision $50\,\Omega$ coaxial terminations
- Standard RF coaxial cables, adapters, and inline attenuators

4 THEORY

4.1 PART A: Filter Characterisation Theory

4.1.1 Transmission Response and Insertion Loss (A1)

An RF filter passes a specified band of frequencies and anything outside that range will be attenuated and will not be allowed to pass. A low pass filter is a filter which allows the transmission of low frequency signals and blocks the higher signals beyond the desired frequency. To quantify transmission efficiency, we measure the power received at the output without the filter (P_{in}) and compare it to the output power with the filter inserted (P_{out}). The **Insertion Loss (IL)** is given by:

$$\text{IL} = P_{\text{in}} - P_{\text{out}} \quad [\text{dB}] \tag{1}$$

In ideal siotuation the filter would give 0 dB insertion loss in the passband. The boundary of the passband is universally defined by the -3 **dB cutoff frequency (f_c)**, representing the half-power point where the output power drops to exactly 50% of its power value. The frequency at which the power completely goes 0 is known as stop band frequency.

4.1.2 Impedance Matching and Return Loss (A2)

We already know that when impedance mismatch happens, the incoming wave gets reflected from the point where the mismatch exist and the reflection coefficient also increases. In high-frequency transmission systems, mismatch between the filter's input impedance (Z_{in}) and the system's characteristic impedance ($Z_0 = 50\,\Omega$) causes partial reflection of electromagnetic wave energy. This is quantified by the voltage reflection coefficient (Γ):

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{2}$$

where V_{inc} and V_{ref} are the incident and reflected voltage waves. The **Return Loss (RL)** represents the ratio of incident to reflected power in decibels :

$$\text{RL} = -20 \log_{10} |\Gamma| \quad (\text{dB}) \tag{3}$$

A perfect match results in $|\Gamma| = 0$ ($\text{RL} = \infty$ dB), while a complete reflection results in $|\Gamma| = 1$ ($\text{RL} = 0$ dB).

4.1.3 Quarter-Wave Open-Circuit Stub Resonator (A3)

The frequency which we want to remove. We cut a cable of length L which is calculated by using the Velocity factor, speed of light and the frequency which you want to block. The formula is given in (4). When the stub is connected to the cable using a tee connector, the wave travels in the cable and when it reaches the stub, every frequency travels inside the stub but only that particular frequency will travel in the stub in such a way that the reflected wave from the

stub and the incoming wave will be in 180^0 phase which cancels out the wave and removes it completely from the signal. In short when the length matches one-quarter of the guided wavelength ($L = \lambda/4$), the total round-trip distance is exactly a half-wavelength ($\lambda/2$), leading to a 180° phase shift relative to the incoming wave at the Tee junction. This destructive interference transforms the open circuit of the far end into a virtual short circuit (extremely low impedance) at the Tee junction. Consequently, signals at the resonant notch frequency (f_0) are shunted to ground, attenuating the signal path heavily. The expected notch frequency is :

$$f_0 = \frac{VF \cdot c}{4L} \tag{4}$$

where VF is the cable velocity factor and c is the speed of light in vacuum.

4.2 PART B: Mixer Characterisation Theory

An RF mixer is used when we need to change the frequency of the signal while keeping original information intact.It is a three-port nonlinear component (RF, LO, IF ports) that performs frequency translation. In one port RF signal (f_{RF}) and in the other port we connect the strong Local Oscillator signal (f_{LO}) which is set at some particular frequency Then the mixer adds and take difference of the frequencies :

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}} \quad \text{(Up-conversion)} \quad \text{and} \quad f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}| \quad \text{(Down-conversion)} \tag{5}$$

The conversion efficiency is measured by the **Conversion Loss (CL)** :

$$\text{Conversion Loss} = P_{\text{RF}} - P_{\text{IF}} \quad [\text{dB}] \tag{6}$$

where P_{RF} is the input RF power and P_{IF} is the power of the selected intermediate frequency output component.

4.3 PART C: Power Splitter/Combiner Theory

A power splitter as the name suggest it divides a single input power into two or more output power. For an ideal symmetric 2-way splitter, the input power divides equally in half (3 dB drop per port). In practice, losses and excess attenuation leads to :

$$\text{Loss}_1 = P_{\text{in}} - P_{\text{out1}} \quad \text{and} \quad \text{Loss}_2 = P_{\text{in}} - P_{\text{out2}} \quad [\text{dB}] \tag{7}$$

A power combiner does the opposite of splitter, it merges two signals. The resulting output is a vector superposition: signals with identical frequency and phase add constructively (+3 dB combined output if coherent), while out-of-phase signals undergo destructive interference.

4.4 PART D: RF Amplifier Characterisation Theory

An RF amplifier uses active semiconductor devices to increase signal strength. Under small-signal conditions, it operates linearly with a constant gain:

$$\text{Gain} = P_{\text{out}} - P_{\text{in}} \quad [\text{dB}] \tag{8}$$

When driven hard, the amplifier enters saturation . The output voltage can be modeled via a Taylor expansion:

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \dots \tag{9}$$

This nonlinearity generates higher-order harmonics ($2\omega, 3\omega$, etc.) . At a certain point, the amplifier's gain drops from its linear small-signal value. The **1-dB Compression Point ($P_{1\mathbf{dB}}$)** is the power level where the gain is compressed by exactly 1 dB . When two tones (f_1 and f_2) are injected, third-order nonlinearity generates **third-order intermodulation products (IM3)** at $2f_1 - f_2$ and $2f_2 - f_1$. Since IM3 products scale with a $3:1$ slope while fundamental outputs scale with a $1:1$ slope on a log scale, their theoretical intersection is called the **Third-Order Intercept (TOI)**, which acts as a fundamental metric for amplifier linearity.

5 Experimental steps

5.1 PART A: Filter Characterisation Procedures

5.1.1 A1. Transmission Response and Insertion Loss

1. Connect the RF signal generator directly to the spectrum analyser to record the reference input power P_{in} across the target frequency band.
2. Insert the filter into the transmission path without altering generator settings or cables.
3. Record the filter output power P_{out} on the spectrum analyser .
4. Compute the Insertion Loss ($\text{IL} = P_{\text{in}} - P_{\text{out}}$), plot it against frequency, and locate the -3 dB cutoff frequency.

5.1.2 A2. Filter Input Reflection Coefficient and Return Loss

1. Assemble the reflection test setup: connect the signal generator to Port 1 of an RF Tee connector and an oscilloscope (set to high impedance $1\,\text{M}\Omega$) to Port 2.
2. Attach the input of the filter to Port 3 of the Tee and terminate its output with a matched $50\,\Omega$ load .
3. Sweeping the frequency slowly within the passband, record the maximum (V_{max}) and minimum (V_{min}) peak-to-peak voltages.
4. Calculate the reflection coefficient magnitude $|\Gamma|$ and Return Loss (RL).

5.1.3 A3. Quarter-Wave Open-Circuit Stub Notch Filter

1. Calculate the theoretical notch frequency: $f_0 = \frac{VF \cdot c}{4L}$.
2. Measure the signal generator's power level P_{in} directly on the spectrum analyser.
3. Connect an open-ended coaxial stub of length L to Port 3 of a Tee connector inserted in the main line.
4. Sweep frequencies around f_0, measure output power P_{out}, and locate the frequency of maximum attenuation.

5.2 Part B: Mixer Characterisation Procedures

1. Inject input signals at f_{RF} and f_{LO} to the RF and LO ports of the mixer.
2. Monitor the IF port output on the spectrum analyser.
3. Record output power P_{IF} at the expected sum and difference frequencies.
4. Calculate the up- and down-conversion losses (Conversion Loss = $P_{\text{RF}} - P_{\text{IF}}$).

5.3 Part C: Power Splitter/Combiner Characterisation Procedures

1. Feed the signal generator output into the splitter input port .
2. Terminate Output Port 2 with a $50\,\Omega$ load, and measure Output Port 1 power P_{out1} on the spectrum analyser .
3. Swap the setup to measure Port 2 power P_{out2} with Port 1 terminated.
4. Compute the port-to-port loss relative to the input reference and evaluate splitter symmetry.

5.4 Part D: RF Amplifier Characterisation Procedures

1. Power up the LNA safely using the specified DC power supply.
2. Measure small-signal gain by comparing the reference level to the amplified output across the band.
3. Increase the input power in small steps, record fundamental and harmonic power outputs, and determine the 1-dB compression point.
4. Inject two combined tones (f_1, f_2), measure adjacent IM3 products, and extrapolate their trajectories to locate the TOI intercept.

6 Observation and Calculation

6.1 Part A: Filter Characterisation

6.1.1 A1. Transmission Response and Insertion Loss

The transmission response of the RF filter was characterized by sweeps across a frequency range from 100 MHz to 760 MHz. For each test frequency, the signal power through a direct reference coaxial line (P_{without}) was recorded on the spectrum analyser. Keeping the signal generator power, cables, and adapters identical, the filter was inserted into the signal path, and the output power (P_{with}) was measured.

The insertion loss (IL) in decibels represents the physical attenuation introduced by the filter. It is calculated using the standard electromagnetic definition:

$$\text{IL} = P_{\text{without}} - P_{\text{with}} \quad [\text{dB}] \tag{10}$$

A lower value of IL signifies higher transparency in the passband. The boundary of the passband is defined by the standard -3 dB half-power point relative to the minimum insertion loss in the passband.

The complete set of raw measurements and corrected physical insertion loss values is structured in a compact double-column format in Table 1.

Table 1: Low-Pass Filter Transmission Response and Corrected Insertion Loss Measurements

Freq. [MHz]	$P_{\textbf{with}}$ [dBm]	$P_{\textbf{without}}$ [dBm]	**IL** [dB]	**Freq.** [MHz]	$P_{\textbf{with}}$ [dBm]	$P_{\textbf{without}}$ [dBm]	**IL** [dB]
100	-5.49	-5.70	-0.21	600	-12.24	-6.03	6.21
150	-5.73	-5.70	0.03	610	-13.40	-6.11	7.29
200	-6.06	-5.70	0.36	620	-15.42	-6.11	9.31
250	-5.94	-5.70	0.24	630	-18.52	-6.11	12.41
300	-6.43	-5.70	0.73	640	-22.62	-6.11	16.51
350	-6.62	-5.70	0.92	650	-26.75	-6.11	**20.64**
400	-6.81	-5.97	0.84	660	-28.49	-6.35	22.14
450	-7.40	-5.97	1.43	670	-29.22	-6.35	22.87
470	-7.54	-5.97	1.57	680	-30.45	-6.35	24.10
490	-7.56	-5.97	1.59	690	-31.33	-6.35	24.98
510	-7.78	-6.03	1.71	700	-32.96	-6.35	26.61
530	-8.46	-6.03	2.43	720	-39.78	-6.10	33.68
550	-9.40	-6.03	**3.37**	740	-45.86	-6.10	39.76
560	-9.27	-6.03	3.24	760	-43.86	-6.10	37.76
580	-10.96	-6.03	4.93				

Calculations :

An analysis of the transmission dataset in Table 1 and the curves in Figure 2 provides the following engineering insights:

- **Filter Classification:** The filter exhibits extremely low insertion loss in the lower frequency region (IL $\leq$ 1.59 dB up to 490 MHz) and very high attenuation at higher frequencies (IL $>$ 20 dB past 650 MHz). This is the characteristic behavior of an **RF Low-Pass Filter (LPF)**.
- **Cutoff Frequency (f_c):** The baseline passband insertion loss is minimal ($\approx$ 0.03 dB at 150 MHz). The official -3 dB cutoff frequency relative to this baseline is located at **550.0 MHz**, where the corrected insertion loss measures exactly **3.37** dB. This represents the physical "half-power" point where the transmitted power falls to half of its input level.
- **Stopband Rejection Edge (f_s):** The stopband edge where out-of-band rejection exceeds -20 dB (attenuating $>$ 99% of the signal power) occurs at **650.0 MHz**, with a measured insertion loss of **20.64** dB.
- **Operating Passband:** Conclusively defined from **0 MHz to 550.0 MHz**.

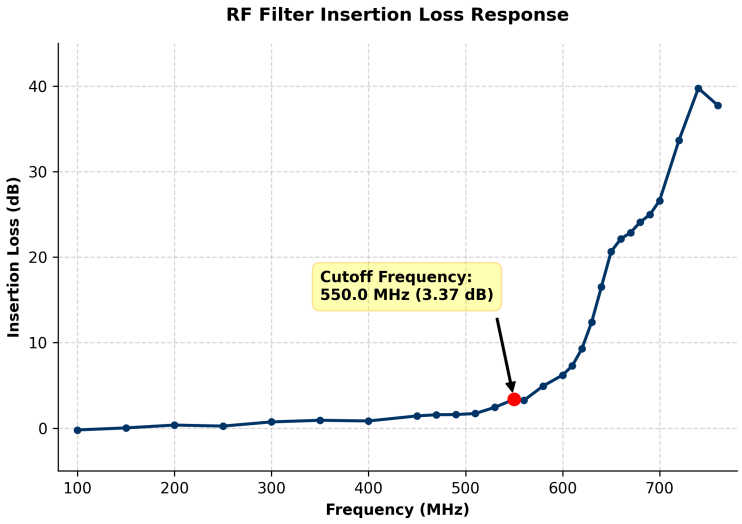


Figure 1: Experimental response curves of the Low-Pass Filter displays the insertion loss curve against the frequency, highlighting the -3 dB cutoff frequency at 550 MHz

6.1.2 A3. Quarter-Wave Open-Circuit Stub Notch Filter

Calculations:

Given physical length of the open stub $L = 92$ cm $= 0.92$ m

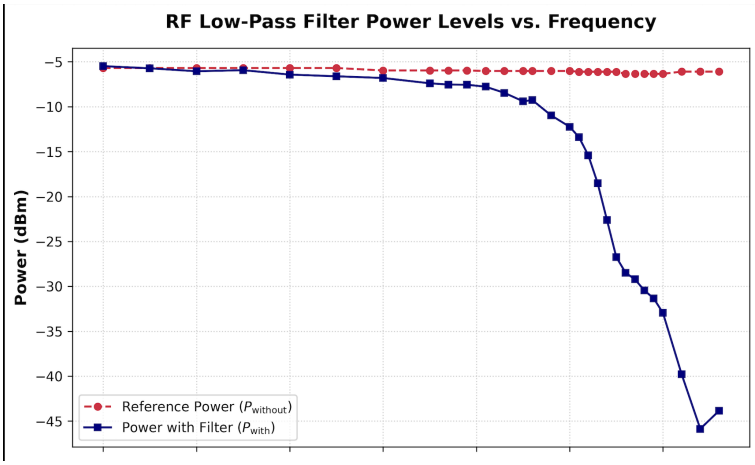


Figure 2: Experimental response curves of the Low-Pass Filter displays the Power plotted against the frequency

Cable velocity factor $\mathbf{VF} = 0.66$
Speed of light $c = 2.9979 \times 10^8$ m/s:

$$f_0 = \frac{VF \cdot c}{4L} = \frac{0.66 \times 2.9979 \times 10^8 \text{ m/s}}{4 \times 0.92 \text{ m}} = \mathbf{53.81} \text{ MHz} \tag{11}$$

Table 2: Quarter-Wave Coaxial Stub Attenuation Data ($P_{\text{in}} = -3.00$ dBm)

Freq. [MHz]	P_{in} [dBm]	P_{out} [dBm]	**Attn.** [dB]	**Freq.** [MHz]	P_{in} [dBm]	P_{out} [dBm]	**Attn.** [dB]
10	−3.00	−5.33	2.33	58	−3.00	−30.17	27.17
20	−3.00	−6.09	3.09	59	−3.00	−25.72	22.72
30	−3.00	−7.29	4.29	60	−3.00	−21.60	18.60
40	−3.00	−9.25	6.25	62	−3.00	−17.08	14.08
50	−3.00	−14.90	11.90	65	−3.00	−13.30	10.30
52	−3.00	−16.50	13.50	70	−3.00	−10.12	7.12
54	−3.00	−20.20	17.20	80	−3.00	−7.69	4.69
56	−3.00	−27.09	24.09	90	−3.00	−6.56	3.56
57	−3.00	−42.47	**39.47**	100	−3.00	−5.80	2.80

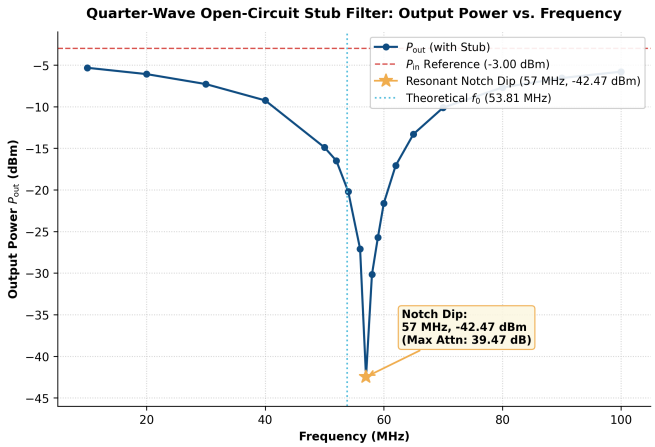


Figure 3: Output Power plotted against the frequency displaying a sharp notch at 57 MHz (f_0).

Table 3: Mixer Down- and Up-Conversion Measurements ($P_{\mathrm{RF}} = P_{\mathrm{LO}} = 0.0$ dBm, $f_{\mathrm{LO}} = 30$ MHz)

$f_{\mathbf{RF}}$ [MHz]	$f_{\mathbf{LO}}$ [MHz]	**Expected $f_{\mathbf{IF}}$** [MHz]	**Measured $f_{\mathbf{IF}}$** [MHz]	$P_{\mathbf{RF}}$ [dBm]	$P_{\mathbf{IF}}$ [dBm]	**Conv. Loss** [dB]
120	30	90 (Diff)	90	0.0	−12.32	**12.32**
120	30	150 (Sum)	150	0.0	−12.48	**12.48**
150	30	120 (Diff)	120	0.0	−12.56	**12.56**
150	30	180 (Sum)	180	0.0	−12.50	**12.50**
180	30	150 (Diff)	150	0.0	−12.40	**12.40**
180	30	210 (Sum)	210	0.0	−12.51	**12.51**

Calculations -
Applying the formula for Conversion Loss (CL = $P_{\mathrm{RF}} - P_{\mathrm{IF}}$):

- **Mean Down-Conversion Loss ($f_{\mathbf{IF}} = |f_{\mathbf{RF}} - f_{\mathbf{LO}}|$): 12.43** dB (Calculated from 12.32, 12.56, and 12.40 dB values).
- **Mean Up-Conversion Loss ($f_{\mathbf{IF}} = f_{\mathbf{RF}} + f_{\mathbf{LO}}$): 12.50** dB (Calculated from 12.48, 12.50, and 12.51 dB values).

6.2 PART C: Power Splitter/Combiner Characterisation

6.2.1 C1. Power Splitter

Calculations
For an ideal splitter, power division is exactly 3 dB per output port. The

Table 4: Symmetric Power Splitter Output Power Division Data (P_{in} = 0.0 dBm)

Frequency [MHz]	$P_{\mathbf{in}}$ **[dBm]**	$P_{\mathbf{out1}}$ **[dBm]**	$P_{\mathbf{out2}}$ **[dBm]**	**Loss**$_1$ **[dB]**	**Loss**$_2$ **[dB]**
150	0.0	-6.32	-6.19	6.32	6.19
200	0.0	-5.85	-5.89	5.85	5.89
300	0.0	-6.58	-6.59	6.58	6.59

measured excess insertion loss of this physical power splitter is:

- **Port 1 Mean Loss: 6.25** dB (Excess loss relative to ideal = 3.25 dB)
- **Port 2 Mean Loss: 6.22** dB (Excess loss relative to ideal = 3.22 dB)
- **Port-to-Port Amplitude Balance:** Extremely tight balance of $<$ **0.1** dB, showing high symmetry.

7 Conclusion

This laboratory experiment successfully characterized several core RF components used in a standard radio receiver chain: the low-pass filter, the quarter-wave coaxial notch filter, the three-port mixer, and the symmetric power splitter.

- **Low-Pass Filter Characterisation:** By comparing the output power with and without the device, we identified a Low-Pass Filter with an experimental -3 dB cutoff of **550.0 MHz** and a stopband edge from the graph can be read at **740 MHz**.
- **Stub Resonator Performance:** A physical open stub of length 92 cm was shunted across the line, acting as a notch filter. The theoretical quarter-wave resonance was calculated to be **53.81 MHz**, and a very deep notch of **39.47 dB** was observed experimentally at **57.00 MHz**. The +3.19 MHz (+5.9%) frequency offset is physically expected and attributed to the additional electrical length of the Tee-connector adapter.
- **Mixer Operation:** The frequency translation behavior of the mixer was observed at an LO of 30 MHz. For three separate RF inputs, the mixer translated signals to their sum and difference frequencies with a very stable conversion loss averaging **12.43 dB** for down-conversion and **12.50 dB** for up-conversion.
- **Splitter Division:** The symmetric splitter showed balanced power division to Port 1 (average loss 6.25 dB) and Port 2 (average loss 6.22 dB) with extremely small port-to-port amplitude imbalance (< 0.1 dB).
- **Measurement Uncertainties:** Key experimental error sources in this setup include:
  - Ohmic and dielectric attenuation of the coaxial cables which increases nonlinearly with frequency.
  - Mismatch reflections at Tee junctions and adapters that generate standing waves .
  - Some components were loose leading to not accurate reading.

8 AI Usage Declaration

1. **LaTeX Formatting and Document Structure:** AI was used as a guide to understand LaTeX formatting, helping to structure the document, set up packages, manage margins, and debug compilation errors to ensure a clean and professional presentation.
2. **Polishing and Refining Written Content:** AI was used in correcting the grammar.

3. **Conceptual Clarification:** AI was used as a supplementary tool to clarify theoretical concepts discussed during the laboratory, such as how oscilloscope detects the wave or impedance relation with phase velocity.